Sanjay Kumar Srikakulam

Germany · GitHub · Google Scholar · ORCID · ResearchGate · LinkedIn · Zenodo

From the atoms of a single protein to petabytes of sequencing data, Sanjay works across both the research and the infrastructure of computational biology. A postdoctoral researcher at Forschungszentrum Jülich, he applies AI and agentic frameworks to the biocuration of mass spectrometry literature, explores small language models (SLMs) for annotating mass spectra, and builds algorithms for large-scale sequence indexing. His earlier work spanned structural bioinformatics and molecular dynamics, and he has run the Galaxy Europe platform. He also works with de.NBI and ELIXIR Germany on research cloud infrastructure.

EDUCATION

PhD, Bioinformatics

2024 · Helmholtz Institute for Pharmaceutical Research Saarland and Saarland University, Saarbrücken, Germany

M.Sc., Bioinformatics

2018 · Saarland University, Saarbrücken, Germany

B.Tech., Bioinformatics

2014 · SRM University, Kancheepuram, India

EXPERIENCE

Postdoctoral Researcher

09/2025 to Present · Forschungszentrum Jülich GmbH, Jülich, Germany

- AI and agentic frameworks for biocuration of metabolomics and lipidomics literature: named entity recognition, ontologies, controlled vocabularies, and knowledge graphs.
- SeqSidecar, a columnar query engine for nucleotide and protein sequencing reads, supporting filtering, search, deduplication, and remote index-based querying of FASTA and FASTQ at scale.
- An OCI container registry that mirrors the full BioContainers catalogue of over 100,000 images, with CVE scanning, SBOM generation, and image signing, run as a European service through de.NBI and ELIXIR.
- The de.NBI service registration platform, where de.NBI and ELIXIR Germany services are registered through a structured form with a REST API, EDAM ontology annotations, bio.tools integration, and an admin review portal.
- A Koina inference service operated and supported on the de.NBI Cloud as part of the federated Koina network for machine learning models in proteomics, reachable independently over HTTP, for example from Python or curl.
- Novel algorithms and data structures for indexing and querying petabyte-scale sequencing datasets.
- Administration, strategic planning, and implementation of de.NBI infrastructure and services, focused on cloud infrastructure, AI, and agentic frameworks.
- Administration and strategic expansion of de.NBI and ELIXIR Germany infrastructure and services.
- Demonstrated EOSC EU Node use cases through the EOSC United project.
- Part of the EOSC Entrust project, establishing blueprints for secure data processing environments.
- Talks, posters, and workshops at national and international conferences.
- Represented de.NBI and ELIXIR Germany in national initiatives (NFDI) and European Commission projects.

Scientific Researcher (Postdoc)

08/2022 to 08/2025 · University of Freiburg, Freiburg, Germany

- Administration and management of Galaxy Europe, supporting over 170,000 users.
- Algorithms and optimisations for meta-scheduling, including reducing carbon footprint through efficient scheduling and distributed remote computing.
- Configuration, deployment, administration, and maintenance of the Galaxy HPC OpenStack cloud using Terraform, Ansible, Jenkins, Rsyslog, and the TIG stack.
- Application development, feature innovation, and research.
- Deliverables and reports for national and EU-funded initiatives.
- Research data management through Galaxy.
- Development and deployment of configurations through GitOps and Jenkins CI/CD.
- Scripting and automation.
- Infrastructure monitoring and performance analysis.
- Troubleshooting, logging, and log analysis.
- Support and documentation.
- Partnering with collaborators on national and international projects.
- Represented the organisation at conferences through exhibition booths, presentations, and scientific and industry discussions.

Bioinformatician

02/2021 to 02/2022 · Saarland University Hospital, Homburg, Germany

- Machine learning and survival analysis on longitudinal COPD patient data to identify biomarkers.
- OMICS data analysis.

System Administrator and DevOps, Bioinformatics IT Core Facility

10/2018 to 03/2023 · Helmholtz Institute for Pharmaceutical Research Saarland, Saarbrücken, Germany

- Setup, design, and establishment of the Bioinformatics IT core facility.
- Installation, configuration, and maintenance of CentOS and Red Hat Linux HPC clusters, servers, desktops, and workstations.
- Containerised web services (Gitea, MediaWiki, WordPress, Docker registry) and infrastructure monitoring through Icinga.
- Site reliability engineering processes.
- Budgeting, purchasing, and inventory.
- Bioinformatics application installation, configuration, deployment, and maintenance.
- Support and documentation for all researchers.
- On-demand compute provisioning, physical and virtual, with network and storage allocation.
- Storage provisioning and administration.
- MariaDB database server administration.
- Ansible and custom RPM packages for configuration, monitoring, and maintenance.
- Configuration of iDRAC/IPMI, DNS, TCP/IP, TLS/SSL, LDAP, Active Directory, SSH, Fail2ban, firewall, AutoFS, and printers.
- Troubleshooting and resolution of hardware and software issues.

System Administrator (Research Assistant)

09/2015 to 03/2022 · German Research Centre for Artificial Intelligence (DFKI), Saarbrücken, Germany

- Development and maintenance of an RPM-based distributed configuration management system.
- Development, configuration, and maintenance of web services including MediaWiki, WordPress, Gitea, and reverse proxies.
- Configuration and maintenance of Jenkins build automation and KVM virtual machines.
- Development and maintenance of installation and configuration routines.
- Configuration and maintenance of network monitoring through Icinga 2.
- Development and maintenance of containerised applications using Docker.
- Scripting and automation.
- Monitoring and maintenance of workstations and servers.
- Atlassian stack administration: Crowd, Jira, Confluence, and Bitbucket.
- Support and documentation for the DFKI-ASR user community and the team.

Project Assistant (Student)

05/2013 to 07/2014 · National Institute for Research in Tuberculosis, ICMR, Chennai, India

- Maintained the Mycobacterium tuberculosis structural database (MtbSD).
- Sequence and structural ensemble analysis of HIV and M. tuberculosis proteins.
- Molecular modelling, drug design, and docking.
- Statistical analysis.

PUBLICATIONS

The Galaxy Community. **Galaxy for accessible, reproducible, and collaborative data analyses: 2026 update**. Nucleic Acids Research, 2026. doi:10.1093/nar/gkag469

Bacon WA, Batut B, Srikakulam SK, Zierep P, Bretaudeau A, Grüning B, Le Corguillé G, Hecht H, Davis JY, Hotz HR, Serrano-Solano B. **Ten common misconceptions about Galaxy (and why they are wrong!)**. PLOS Computational Biology, 2026. doi:10.1371/journal.pcbi.1013869

Twardziok S, Barysch SV, Braun M, Buchhalter I, Grüning B, Kraft P, Krüger J, Kuchenbecker L, Srikakulam SK, Lawerenz C, Sczyrba A, Wagener H, Hoffmann N, Kohlbacher O, German Network for Bioinformatics Infrastructure - de.NBI. **Towards Federated, Certified Infrastructures for Sensitive Data Research in Germany and Europe with the de.NBI Cloud**, 2026. doi:10.5281/zenodo.19882942

Srikakulam SK. **Data Snack: Galaxy, a research data powerhouse from analysis to FAIR management**, 2025. doi:10.5281/zenodo.15362960

Srikakulam SK, Azab A, De Geest P, Vondrák T, Kuntz M, Luna-Valero S, Grüning B. **Dynamic meta-scheduling in Galaxy with TPV Broker for smarter workload distribution**, 2025. doi:10.5281/zenodo.16312227

Zierep P, Kuntz M, Srikakulam SK, Grüning B. **Interactive Galaxy Tools: Combining real-time user interaction with scientific workflows**, 2025. doi:10.5281/zenodo.16735306

Azab A, Geest PD, Srikakulam SK, Vondrák T, Kuntz M, Grüning B. **Optimized Meta-Scheduling in Galaxy Using TPV Broker**. ICT for Intelligent Systems (ICTIS 2025), Smart Innovation, Systems and Technologies, Springer, 2025. doi:10.1007/978-981-95-1365-9_42

Azab A, Srikakulam SK, De Geest P, Vondrák T, Grüning B, Kuntz M, Fernandez-del-Castillo E, Luna-Valero S. **Publication on the smart job scheduler implementation**, 2025. doi:10.5281/zenodo.15827107

Tangaro MA, Nicotri S, Grüning B, Srikakulam SK, Dadras A, Kaiser O, Kuntz M, Bretaudeau A, De Geest P, Luna-Valero S, Chavero Díez M, Fernández González JM, Capella-Gutierrez S, Gelpí JL, Astalos J, Jurič B, Ruda M, Opioła Ł, Bayındır H, GIOIOSA S, De Sanctis G, Zambelli F. **Scaling Scientific Workflows in Europe: Architecture and Deployment of the Galaxy-Pulsar Computational Network**, 2025. doi:10.5281/zenodo.16761933

Braun M, Kanitz A, Krüger J, Kraatz L, Miranda J, Schelp C, Schneider-Lunitz V, Srikakulam SK, Stiensmeier X, Hoffmann N, Twardziok S. **Secure Processing Environments as a Service in the de.NBI Cloud**. BioHackrXiv, 2025. doi:10.37044/osf.io/pzmue_v1

Pedersen M, Srikakulam SK, De Geest P, Fernandez-del-Castillo E, Cristofori A, Luna-Valero S, Tangaro MA, Nicotri S. **Bring Your Own Infrastructure**, 2024. doi:10.5281/zenodo.15729502

Nicotri S, Tangaro MA, Zambelli F, Ruda M, Krenek A, Grüning B, Srikakulam SK, Bretaudeau A, Batalden S, Chavero Díez M, De Geest P. **D3.1 Operations documentation on the Open Infrastructure deployment**, 2024. doi:10.5281/zenodo.15920451

The Galaxy Community. **The Galaxy platform for accessible, reproducible, and collaborative data analyses: 2024 update**. Nucleic Acids Research, 2024. doi:10.1093/nar/gkae410

Zierep P, Srikakulam SK, Schaaf S, Grüning B. **Conda, Container and Bots: How to Build and Maintain Tool Dependencies in Workflows and Training Materials**. Proceedings of the Conference on Research Data Infrastructure, 2023. doi:10.52825/cordi.v1i.417

Kayikcioglu T, Srikakulam SK, Maier W, Grüning B. **Interactive Tools (IT) in Galaxy: Combining Synchronous and Asynchronous Workflows**, 2023. doi:10.5281/zenodo.8335099

Srikakulam SK, Keller S, Dabbaghie F, Bals R, Kalinina OV. **MetaProFi: an ultrafast chunked Bloom filter for storing and querying protein and nucleotide sequence data for accurate identification of functionally relevant genetic variants**. Bioinformatics, 2023. doi:10.1093/bioinformatics/btad101

Dabbaghie F, Srikakulam SK, Marschall T, Kalinina OV. **PanPA: generation and alignment of panproteome graphs**. Bioinformatics Advances, 2023. doi:10.1093/bioadv/vbad167

Gress A, Srikakulam SK, Keller S, Ramensky V, Kalinina OV. **d-StructMAn: Containerized structural annotation on the scale from genetic variants to whole proteomes**. GigaScience, 2022. doi:10.1093/gigascience/giac086

Herr C, Mang S, Mozafari B, Guenther K, Speer T, Seibert M, Srikakulam SK, Beisswenger C, Ritzmann F, Keller A, Mueller R, Smola S, Eisinger D, Zemlin M, Danziger G, Volk T, Hoersch S, Krawczyk M, Lammert F, Adams T, Wagenpfeil G, Kindermann M, Marcu C, Ataya ZWD, Mittag M, Schwarzkopf K, Custodis F, Grandt D, Schaefer H, Eltges K, Lepper PM, Bals R. **Distinct Patterns of Blood Cytokines Beyond a Cytokine Storm Predict Mortality in COVID-19**. Journal of Inflammation Research, 2021. doi:10.2147/JIR.S320685

Dultz G, Srikakulam SK, Konetschnik M, Shimakami T, Doncheva NT, Dietz J, Sarrazin C, Biondi RM, Zeuzem S, Tampé R, Kalinina OV, Welsch C. **Epistatic interactions promote persistence of NS3-Q80K in HCV infection by compensating for protein folding instability**. Journal of Biological Chemistry, 2021. doi:10.1016/j.jbc.2021.101031

Richter R, Kamal MAM, García-Rivera MA, Kaspar J, Junk M, Elgaher WAM, Srikakulam SK, Gress A, Beckmann A, Grißmer A, Meier C, Vielhaber M, Kalinina O, Hirsch AKH, Hartmann RW, Brönstrup M, Schneider-Daum N, Lehr CM. **A hydrogel-based in vitro assay for the fast prediction of antibiotic accumulation in Gram-negative bacteria**. Materials Today Bio, 2020. doi:10.1016/j.mtbio.2020.100084

Srikakulam SK, Bastys T, Kalinina OV. **A shift of dynamic equilibrium between the KIT active and inactive states causes drug resistance**. Proteins: Structure, Function, and Bioinformatics, 2020. doi:10.1002/prot.25963

Hassan S, Srikakulam SK, Chandramohan Y, Thangam M, Muthukumar S, Gayathri Devi PK, Hanna LE. **Exploring the conformational landscapes of HIV protease structural ensembles using principal component analysis**. Proteins: Structure, Function, and Bioinformatics, 2018. doi:10.1002/prot.25534

SKILLS

AI and LLM: vLLM, PydanticAI, Coding agents, OpenAI API, RAG, Embeddings, NER, Hugging Face Transformers, pgvector

Bioinformatics methods: Sequence analysis and indexing, Structural bioinformatics, Structural annotation, Molecular modelling, Molecular dynamics, Molecular docking, Virtual screening, Ontology and controlled vocabularies

CI/CD and version control: Git, Jenkins

Cloud, containers and virtualisation: OpenStack, AWS, Docker, LXC, KVM

Data science and ML: pandas, NumPy, scikit-learn

Databases: SQL, MySQL/MariaDB, PostgreSQL

Infrastructure as code and configuration: Terraform, Ansible, RPM packaging, Pulp

Monitoring and observability: Icinga, Telegraf, InfluxDB, Grafana

Networking, DNS and storage: AWS Route 53, DigitalOcean, NAS, DAS, JBOD, NFS, S3, CVMFS

Programming: Python, Bash, R

Security: Firewall management, SSL/TLS certificates, Let's Encrypt, Fail2ban

Web servers: Apache, Nginx

Workload scheduling: HTCondor